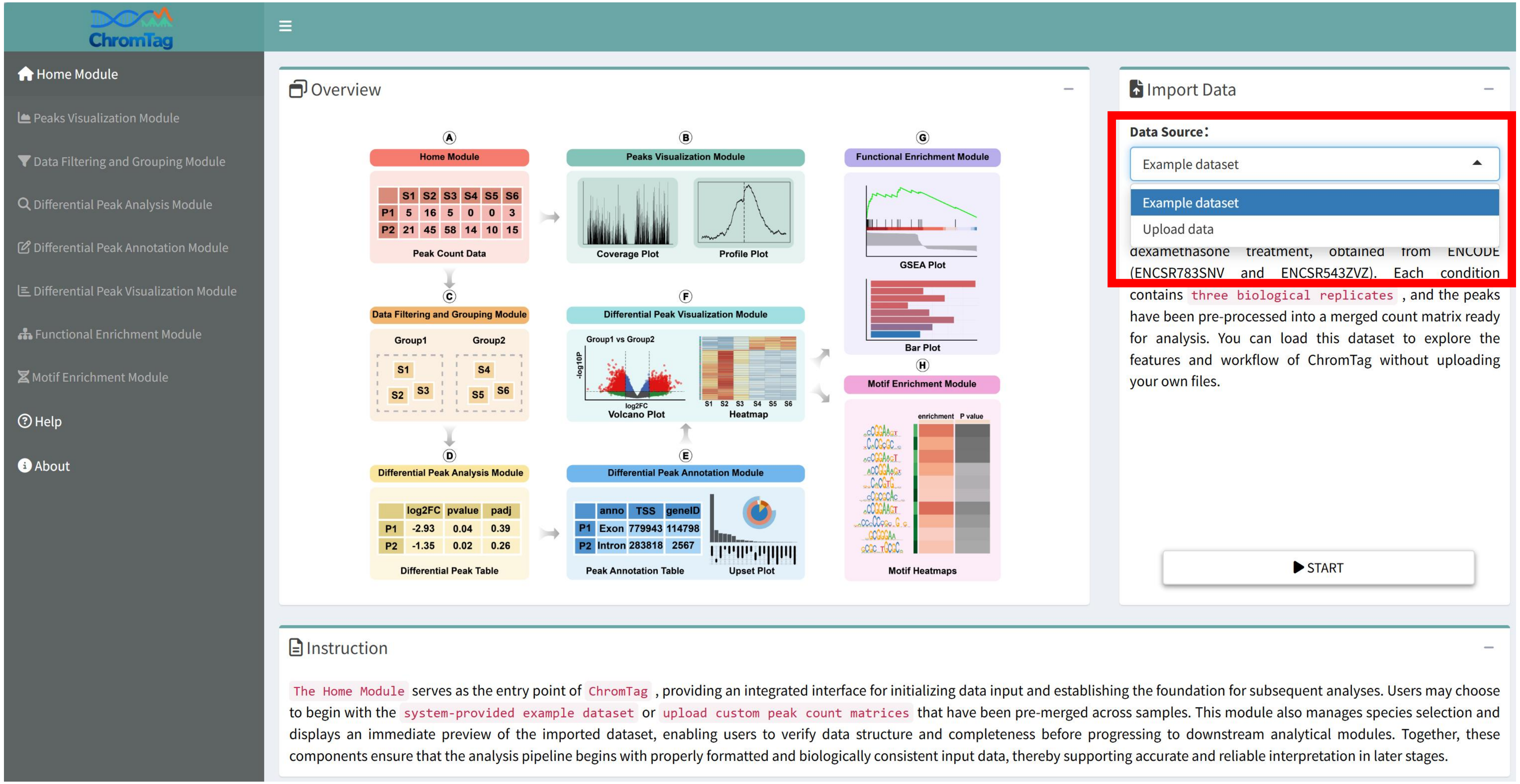
ChromTag

Basic Tutorial

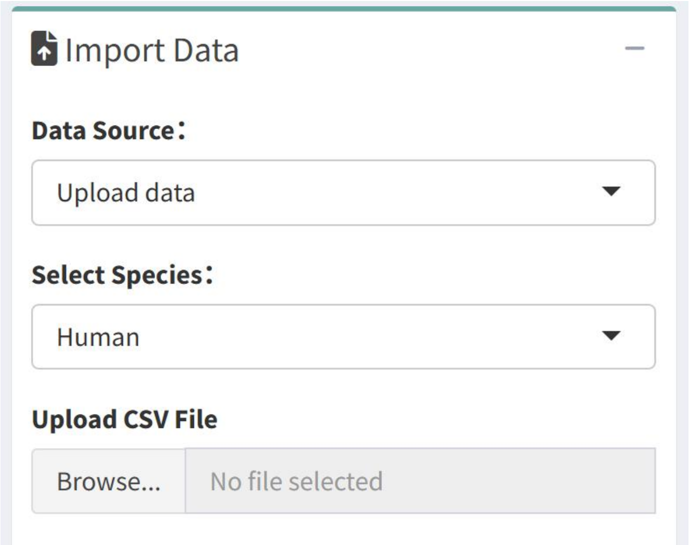
Home Module



Users can choose between **“Example Dataset”** and **“Upload Data”**.

If the **Example Dataset** is selected, the application automatically loads a **pre-processed merged peak count matrix** containing H3K27ac ChIP-seq profiles from A549 cells collected at 0 h and 4 h after dexamethasone treatment. The dataset was obtained from ENCODE, including ENCSR783SNV and ENCSR543ZVZ, with **three biological replicates** for each condition.

Home Module



If you choose to upload custom data, you must also select the appropriate species.

Chromosome	Start	End	A549_0h_rep1	A549_0h_rep2	A549_0h_rep3	A549_4h_rep1	A549_4h_rep2	A549_4h_rep3
chr1	629727	630151	31	12	5	19	18	23
chr1	633792	634272	44	24	10	17	48	25
chr1	778021	778694	46	54	59	69	26	64
chr1	778877	779438	69	53	57	66	37	70
chr1	779666	779906	20	18	22	23	12	29
chr1	826707	827761	97	90	94	88	47	100
chr1	870010	870297	9	17	4	12	6	12
chr1	904028	905405	205	231	182	178	107	185
chr1	906957	907195	28	31	38	18	13	24

Uploaded data should be **pre-merged across samples** and must follow a specific format. The file should be a **CSV** with a header row.

The first column must contain the chromosome name (e.g., “chr1”), the second and third columns should specify the **start** and **end** positions, and all subsequent columns should represent the **peak count values** for each sample.

Each experimental group must include **at least two biological replicates** to ensure proper statistical analysis.

Home Module

Import Data

Data Source:

Example dataset

Example dataset

Upload data

dexamethasone treatment, obtained from ENCODE (ENCSR783SNV and ENCSR543ZVZ). Each condition contains `three biological replicates` , and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

START

Data Preview

Show 10 **entries** **Search:**

	Chromosome	Start	End	A549_0h_rep1	A549_0h_rep2	A549_0h_rep3	A549_4h_rep1	A549_4h_rep2	A549_4h_rep3
1	chr1	629727	630151	31	12	5	19	18	23
2	chr1	633792	634272	44	24	10	17	48	25
3	chr1	778021	778694	46	54	59	69	26	64
4	chr1	778877	779438	69	53	57	66	37	70
5	chr1	779666	779906	20	18	22	23	12	29
6	chr1	826707	827761	97	90	94	88	47	100
7	chr1	870010	870297	9	17	4	12	6	12
8	chr1	904028	905405	205	231	182	178	107	185
9	chr1	906957	907195	28	31	38	18	13	24
10	chr1	910946	911195	18	17	15	13	6	21

Showing 1 to 10 of 76,176 entries

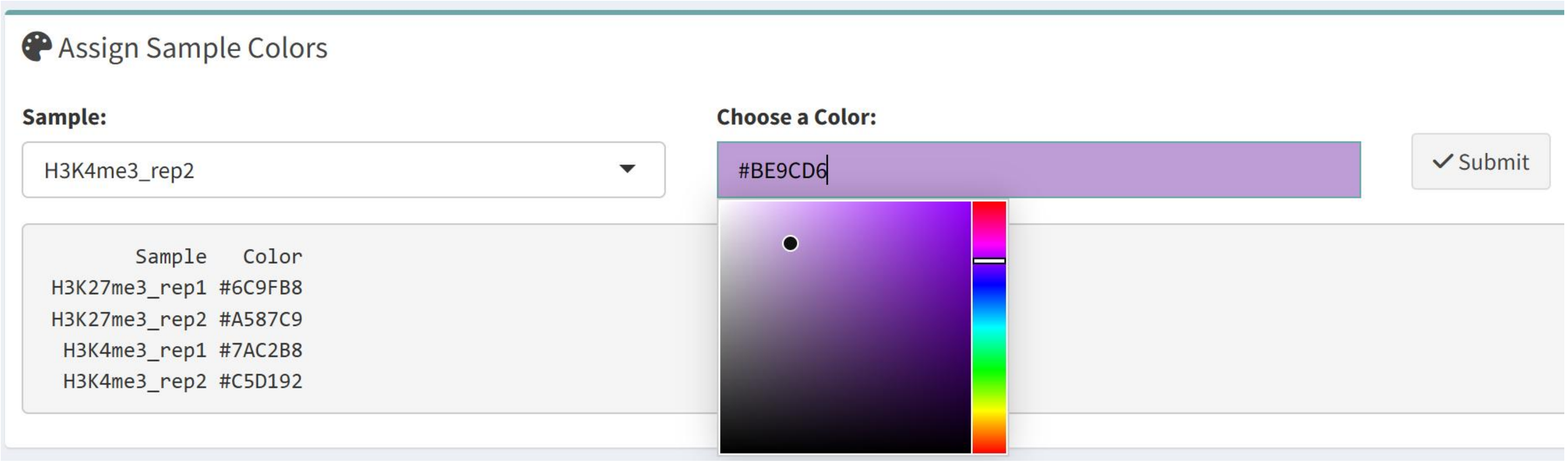
Previous 1 2 3 4 5 … 7,618 Next

After choosing the data source, users can click the **"Start"** button to load the dataset and proceed to the next module.

Once the data is successfully loaded, the system will display the **peak count matrix** in a preview table for inspection.

Peaks Visualization Module

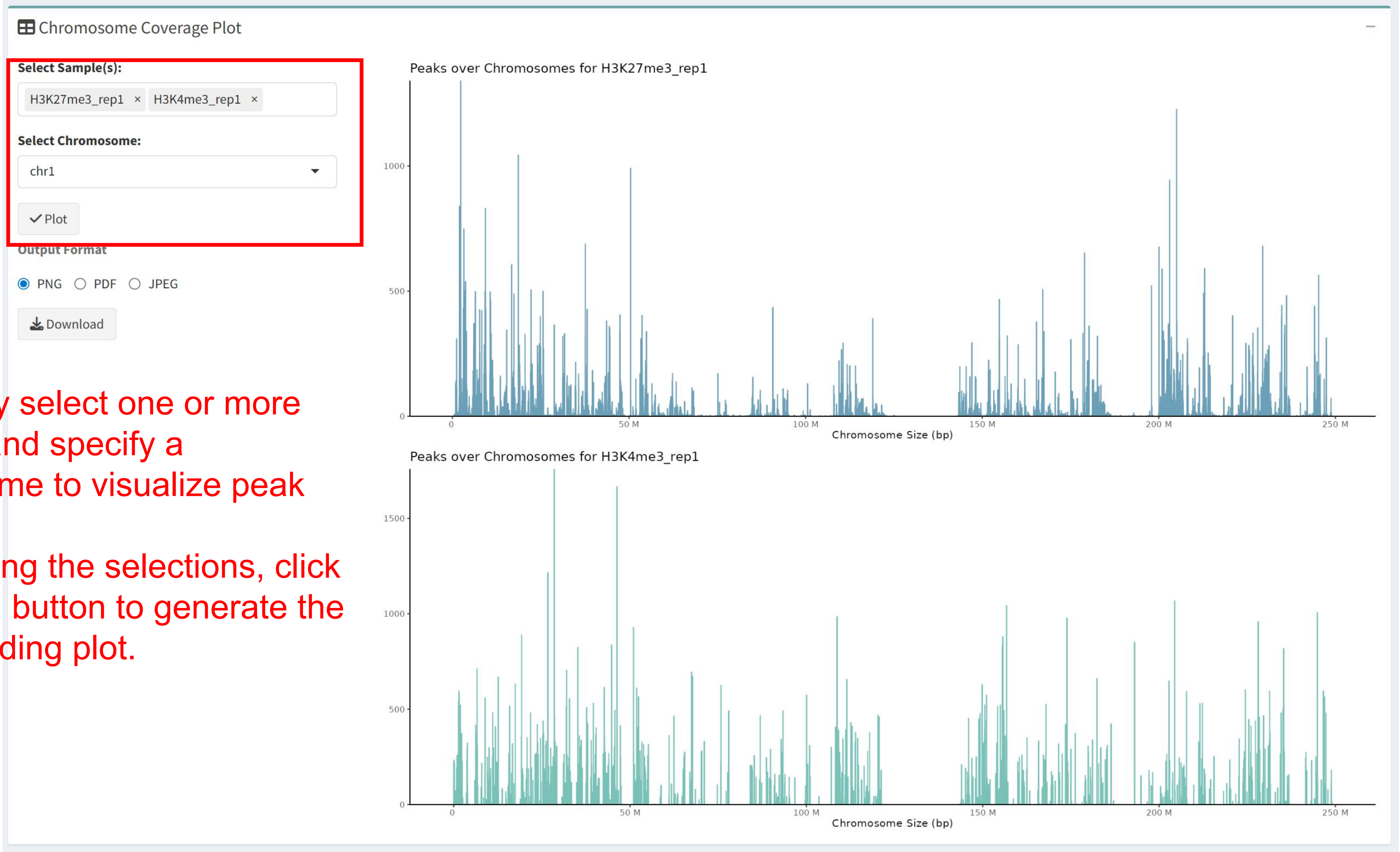
This step is optional and can be skipped.



Users can select a sample and assign a preferred color using the color picker.
Once submitted, the chosen color will be applied consistently across the **Chromosome Coverage Plot** and the **TSS Profile Plot**, enhancing visual distinction among samples.

Peaks Visualization Module

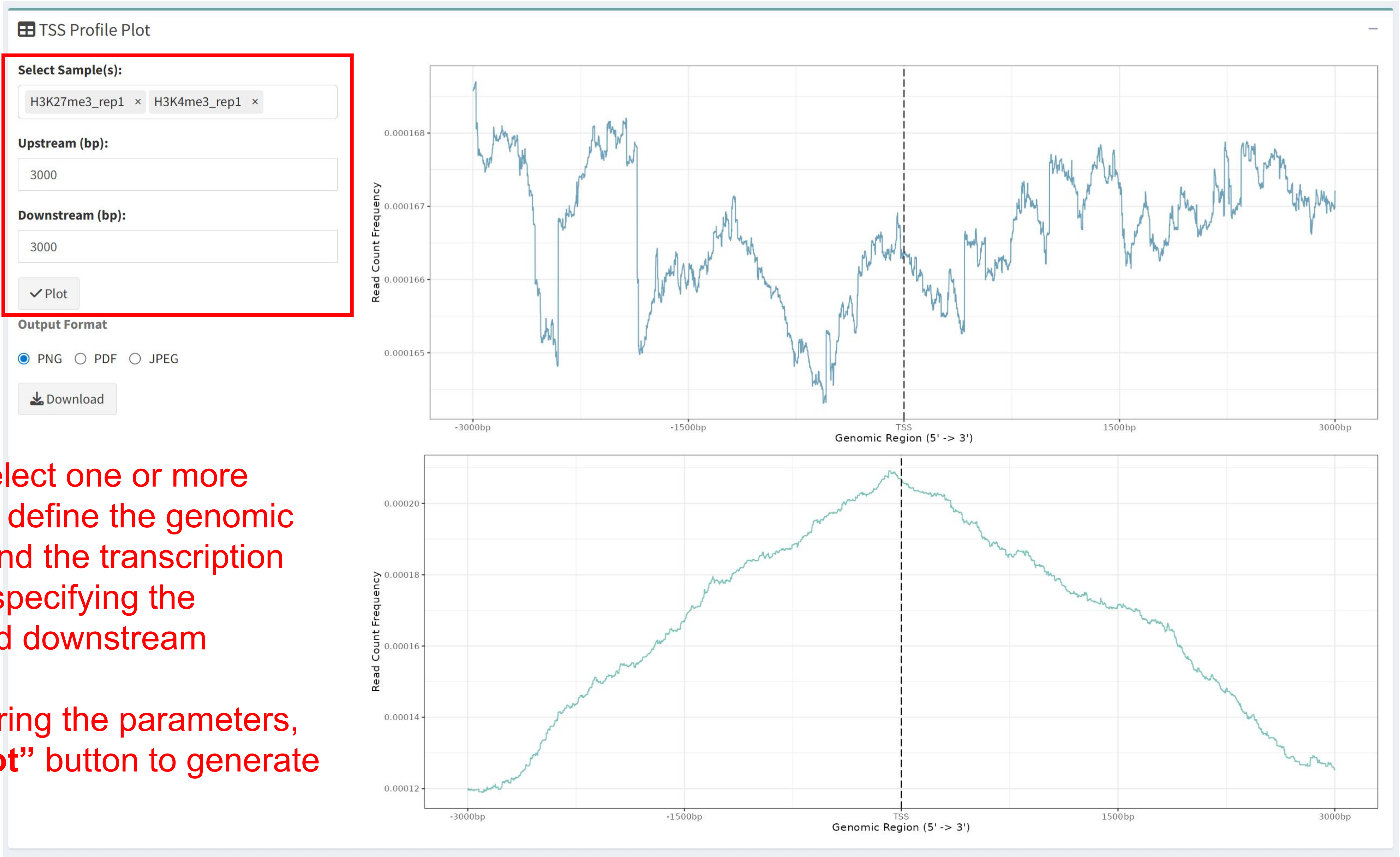
This step is optional and can be skipped.



Users may select one or more samples and specify a chromosome to visualize peak coverage.
After making the selections, click the “**Plot**” button to generate the corresponding plot.

Peaks Visualization Module

This step is optional and can be skipped.



Users can select one or more samples and define the genomic window around the transcription start site by specifying the upstream and downstream distances.

After configuring the parameters, click the **"Plot"** button to generate the plot.

Data Filtering and Grouping Module

Before Filter | After Filter

Count Threshold: ?

5

✓ Submit

Users may specify a threshold value to filter the data.
During filtering, the system sums the read counts for each genomic region across samples and retains only those entries whose total count exceeds the defined threshold.

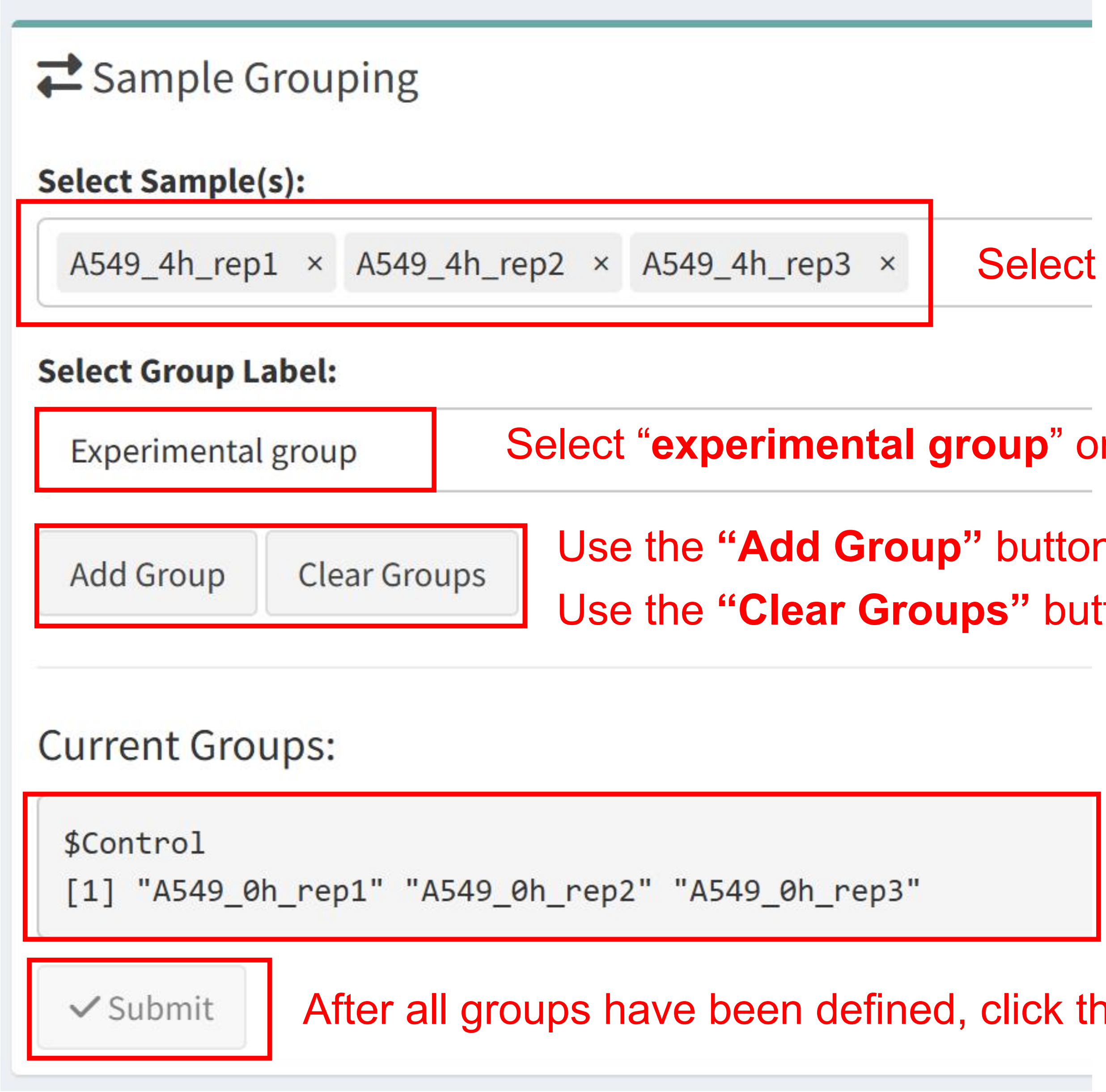
Show 5 **entries** **Search:**

hromosome	Start	End	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_rep1	H3K4me3_rep2	
ır1	27805	30656	0 +	0 +	233 +	186	> 5
ır1	135727	140083	0	2	166	89	
ır1	198335	200968	0	0	219	169	
ır1	391014	392146	1	3	0	1	
ır1	392617	393317	2 +	1 +	0 +	0	< 5

Showing 1 to 5 of 220,230 entries

Previous 1 2 3 4 5 … 44,046 Next

Data Filtering and Grouping Module



Select the samples that should be grouped together.

Select “**experimental group**” or “**control group**” as the group name.

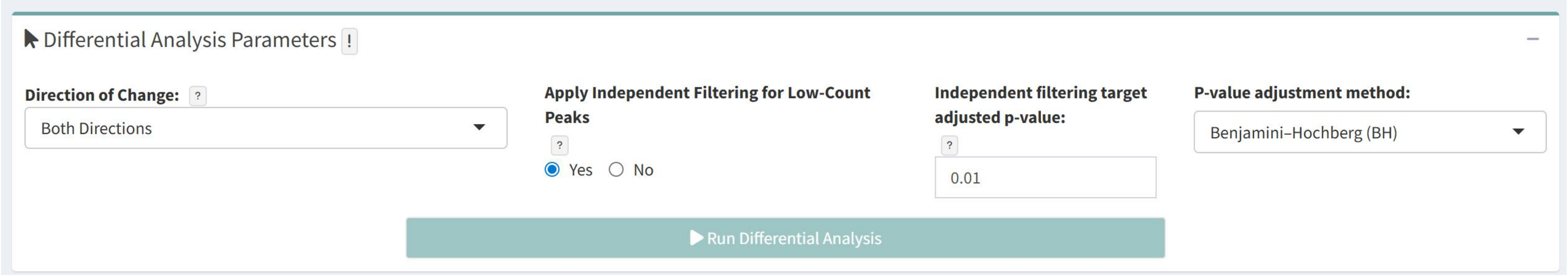
Use the **“Add Group”** button to create the group.

Use the **“Clear Groups”** button to remove all existing groups.

All defined groups are listed under **“Current Groups.”**

After all groups have been defined, click the **“Submit”** button to save the grouping information.

Differential Peak Analysis Module



Direction of Change

This setting lets users choose whether to detect peaks that are increased or decreased between groups, allowing the analysis to focus on a specific direction of change.

Independent Filtering

Independent filtering is a DESeq2 step that ignores low-signal peaks during multiple-testing correction to improve detection power.
Filtered peaks remain in the results, but their adjusted p-values may be NA.

target adjusted p-value

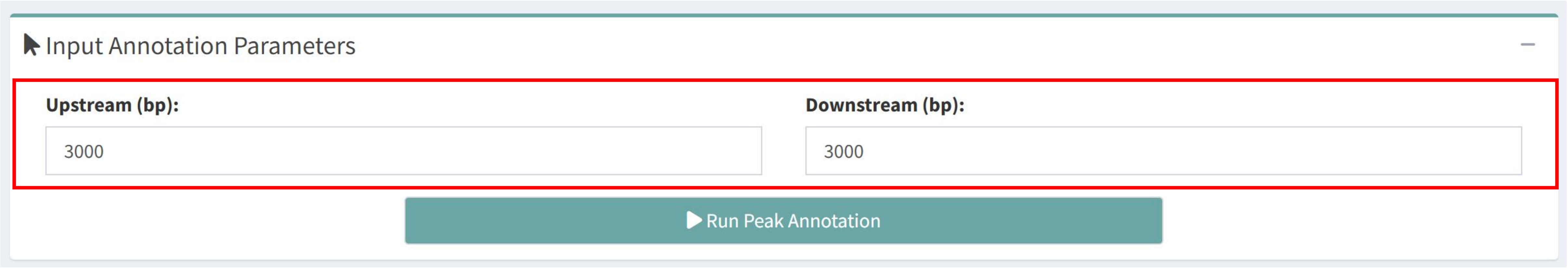
DESeq2 uses this value to choose a mean-count filtering threshold that helps maximize the number of significant peaks after multiple-testing correction. It does not directly define the final differential peaks; final peak selection is still based on the adjusted p-value and log2FC cutoffs specified later.

p-value Adjustment Method

This option allows users to select a method for multiple-testing correction.

- The Benjamini–Hochberg (BH) method controls the false discovery rate (FDR) and is well suited for analyses with many tests.
- The Holm method controls the family-wise error rate (FWER) and is more conservative, reducing false positives but potentially missing true signals.

Differential Peak Annotation Module



To annotate the peaks, first specify the upstream and downstream distances to define the genomic region of interest, and then click the “Run Peak Annotation” button to execute the annotation.

Differential Peak Visualization Module

Volcano Plot | MA Plot | PCA Plot | Heatmap

–log10(padj) Cutoff

1.3

Log2FC Threshold (Positive)

1

Log2FC Threshold (Negative)

1

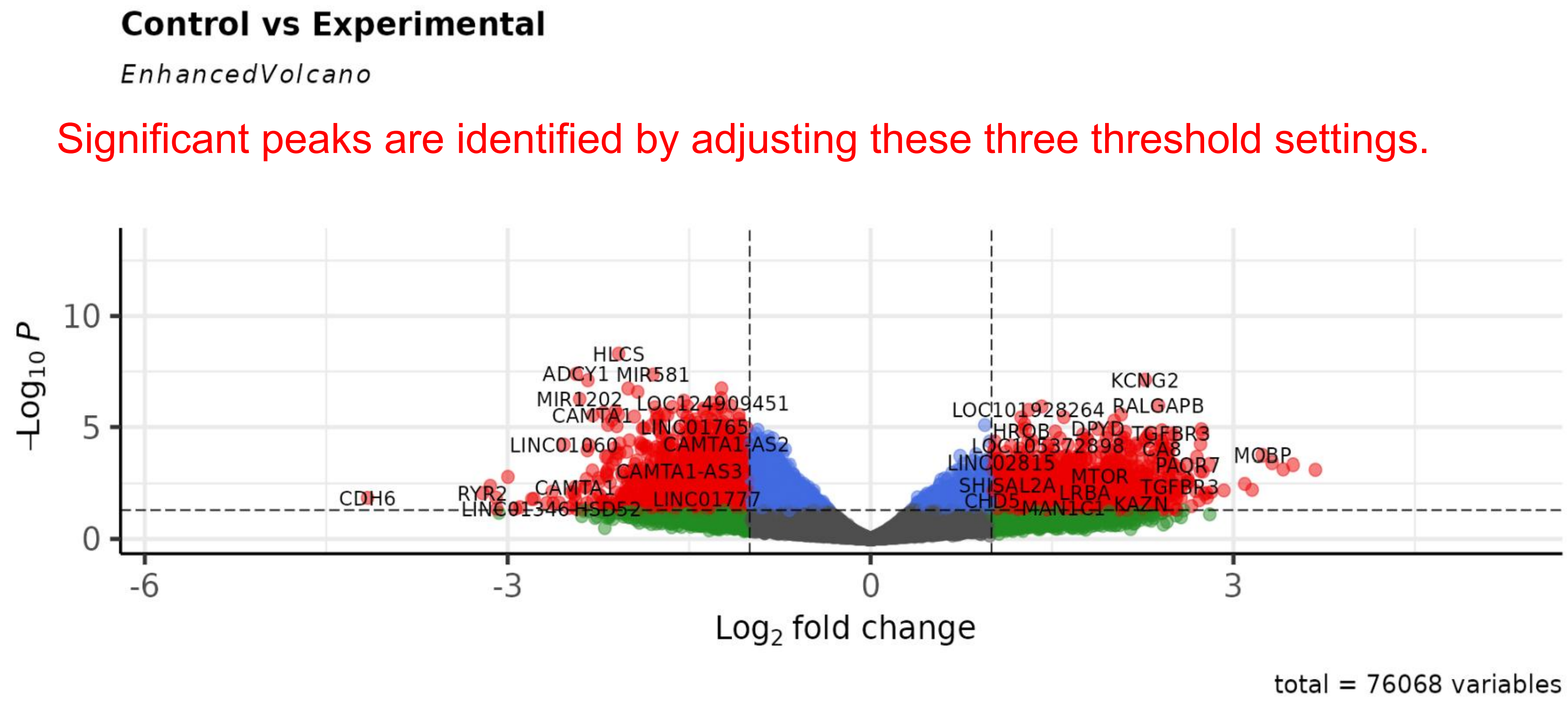
Point Size

3

Label Size

4

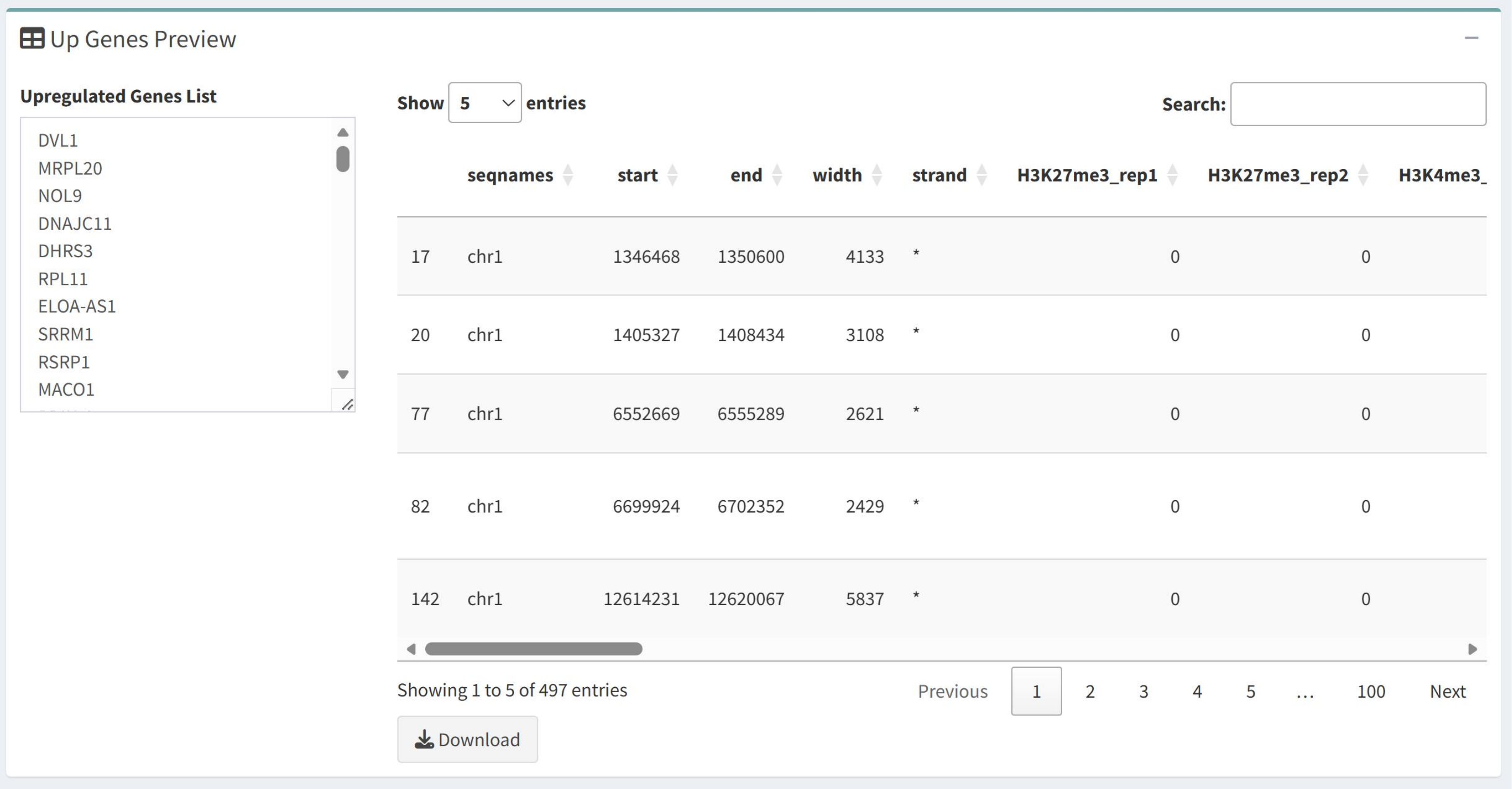
✓ Submit



Significant peaks are identified by adjusting these three threshold settings.

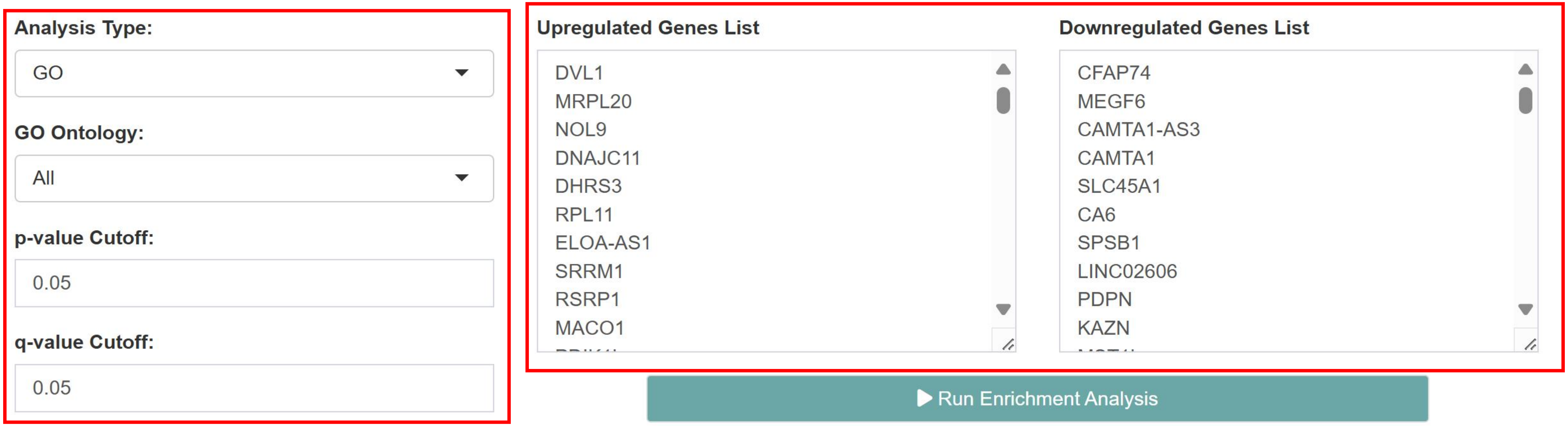
Click the “Submit” button to update the plot and apply the selected significance criteria.

Differential Peak Visualization Module



This section displays the upregulated and downregulated gene lists, as well as the peak tables generated from the filtering step.

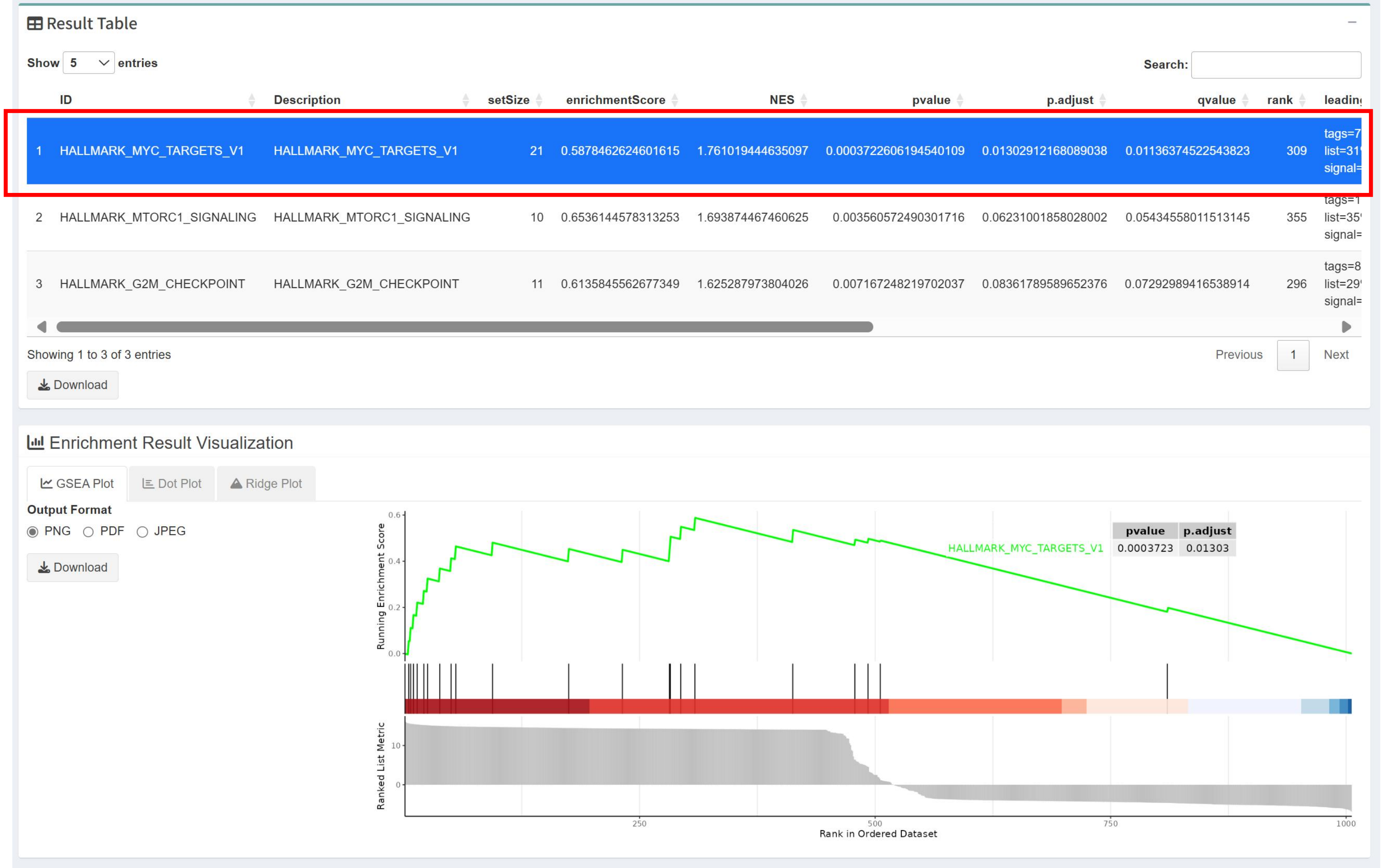
Functional Enrichment Module



Adjust the analysis type and corresponding parameters as needed.

For GO or KEGG analysis, users may enter or edit gene lists using gene symbols to define the input for enrichment testing. When GSEA is selected, the input is limited to the full set of genes annotated in the Gene Annotation module.

Functional Enrichment Module



When GSEA is selected, users must choose a specific enrichment entry before generating the corresponding GSEA plot.

Motif Enrichment Module

Upregulated Peaks List

chr3-150407290-150415877
chr16-2150741-2157314
chr16-88972494-88978084
chr6-119343897-119351533
chr12-56682168-56690455
chr8-38173859-38178882
chr9-128688198-128692391
chr11-65420641-65428391
chr10-92688412-92694886
chr6-27889944-27897392

Downregulated Peaks List

chr7-154746354-154754198
chr11-20155235-20165969
chr7-27100472-27112603
chr1-208072367-208087339
chr1-8323134-8331215
chr11-118150179-118161808
chr20-47665285-47675999
chr14-64734713-64750256
chr1-28810214-28820062
chr4-182814697-182830407

This section displays the top 200 upregulated and 200 downregulated peaks, which are selected from the differential peaks identified through the volcano plot filtering step. You can also input and modify your own lists of upregulated and downregulated peaks for further analysis.

Number of Top Transcription Factors to Plot

10

Display Motif GC Content

- (•) Yes
- ( ) No

Enable Clustering

- (•) Yes
- ( ) No

·**Number of Top Transcription Factors to Plot**: Set how many top transcription factors you want to visualize (default is 10).
·**Display Motif GC Content**: Choose whether to show the GC content of motifs.
·**Enable Clustering**: Decide whether to enable clustering for grouping similar motifs in the plot.